

5. 2

Use mathematical induction in Exercises 3–17 to prove summation formulae. Be sure to identify where you use the inductive hypothesis.

3. Let $P(n)$ be the statement that $1^2 + 2^2 + \cdots + n^2 = n(n+1)(2n+1)/6$ for the positive integer n .
- What is the statement $P(1)$?
 - Show that $P(1)$ is true, completing the basis step of a proof that $P(n)$ is true for all positive integers n .
 - What is the inductive hypothesis of a proof that $P(n)$ is true for all positive integers n ?
 - What do you need to prove in the inductive step of a proof that $P(n)$ is true for all positive integers n ?
 - Complete the inductive step of a proof that $P(n)$ is true for all positive integers n , identifying where you use the inductive hypothesis.
 - Explain why these steps show that this formula is true whenever n is a positive integer.

a) $P(1) = 1^2 = (1 \cdot 1)(1+1)/6$

b) $1^2 = 1$ $((1 \cdot 1)(1+1))/6 = (1 \cdot 2)/6 = 2/6 = 1/3$

c) $P(k)$ is true for all $k \in \mathbb{Z}^+$

d) $P(k+1)$ is true

e) assume $P(k)$ is true

$$P(k+1) = 1^2 + 2^2 + \dots + k^2 + (k+1)^2 = \frac{(k+1)(k+2)(2(k+1)+1)}{6}$$

$$\frac{k(k+1)(2k+1)}{6} + (k+1)^2 = \frac{(k+1)(k+2)(2(k+1)+1)}{6}$$

expand everything, we find LHS = RHS

f) since we've shown $P(k)$ is true, and $P(k+1)$ is true,

we can express as $k+1$, and therefore $P(n)$ is true for all $\mathbb{Z}^+$

6. Prove that $1 \cdot 1! + 2 \cdot 2! + \cdots + n \cdot n! = (n+1)! - 1$ whenever n is a positive integer.

base case 1

$$1 \cdot 1! = (1+1)! - 1 = 2! - 1 = 2 - 1 = 1$$

for an arb. $k \in \mathbb{Z}^+$ assume $P(k) \equiv T$

$$P(k+1) = 1 \cdot 1! + 2 \cdot 2! + \dots + k \cdot k! + (k+1) \cdot (k+1)! = (k+2)! - 1$$

as $1 \cdot 1! + 2 \cdot 2! + \dots + k \cdot k! = (k+1)! - 1$

$$(k+1)! - 1 + (k+1) \cdot (k+1)! = (k+2)! - 1$$

$$(k+1)! + (k+1) \cdot (k+1)! = (k+2)!$$

$$(k+1)! + (k+1) \cdot (k+1)! = (k+2) \cdot (k+1)!$$

$$\frac{(k+1)! \cdot (1 + k + 1)}{(k+2)!} = \frac{(k+2) \cdot (k+1)!}{(k+2)!}$$

$$1 + k + 1 = k + 2$$

thus we show the sequence works for all $\mathbb{Z}^+$

11. a) Find a formula for

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots + \frac{1}{2^n}$$

by examining the values of this expression for small values of n .

b) Prove the formula you conjectured in part (a).

a) Let the formula be denoted as $P(n)$ where $n \in \mathbb{Z}^+$

$$P(1) = \frac{1}{2}, P(2) = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}, P(3) = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} = \frac{7}{8}$$

$$P(n) = \frac{2^n - 1}{2^n}$$

b)

we use it (show in part a)

assume $P(k)$ is true where $k \in \mathbb{Z}^+$ our induction

$$P(k+1) = \frac{2^{k+1} - 1}{2^{k+1}} = \frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2^k} + \frac{1}{2^{k+1}}$$

$$\text{as } \frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2^k} = \frac{2^k - 1}{2^k}$$

$$\frac{2^{k+1} - 1}{2^{k+1}} = \frac{2^k - 1}{2^k} + \frac{1}{2^{k+1}}$$

$$\frac{2 \cdot 2^k - 1}{2 \cdot 2^k} = \frac{2^k - 1}{2^k} \left(\frac{2}{2} \right) + \frac{1}{2 \cdot 2^k}$$

$$\frac{2 \cdot 2^k - 1}{2 \cdot 2^k} = \frac{2(2^k - 1) + 1}{2 \cdot 2^k}$$

$$\frac{2 \cdot 2^k - 1}{2 \cdot 2^k} = \frac{2 \cdot 2^k - 1}{2 \cdot 2^k}$$

thus LHS = RHS

a) the formula $P(k)$ is proven for all $k \in \mathbb{Z}^+$

21. Prove that $2^n > n^2$ if n is an integer greater than 4.

base case $n=5$

$$2^5 = 32$$

$$5^2 = 25$$

$$32 > 25$$

$$P(5) = 2^5 > 5^2$$

show $P(k)$ is true where k is any $\mathbb{Z}^+$ and $k \geq 5$

$P(k+1)$

$$2^{k+1} > (k+1)^2$$

$$2 \cdot 2^k > (k+1)^2$$

$$\text{as } 2^k > k^2$$

$$\begin{aligned}
 2k^2 &> 2k^2 \\
 5k &> 2k > (k+1) \\
 2k &> k+1 \\
 k &> 2k+1 \\
 k^2 &> 0 \\
 (k+1)^2 &> 0 \\
 \text{etc.} \\
 4^2 - 1 &= 15 > 0 \\
 \text{etc.}
 \end{aligned}$$

32. Prove that 3 divides $n^3 + 2n$ whenever n is a positive integer.

$$p(k) = 3 \mid n^3 + 2n$$

$$\begin{aligned}
 p(n) &= 3 \mid n^3 + 2n \\
 p(1) &= 3 \mid 1^3 + 2 \cdot 1 = 3
 \end{aligned}$$

$$\begin{aligned}
 p(k) &= 3 \mid (k+1)^3 + 2(k+1) \\
 p(k+1) &= 3 \mid (k+1)^3 + 2(k+1)
 \end{aligned}$$

$$\begin{aligned}
 (k+1)^3 &= k^3 + 3k^2 + 3k + 1 \\
 (k+1)^3 + 2(k+1) &= k^3 + 3k^2 + 3k + 1 + 2k + 2 \\
 &= k^3 + 3k^2 + 5k + 3
 \end{aligned}$$

$$\begin{aligned}
 &= k^3 + 3k^2 + 5k + 3 \\
 &= k^3 + 3k^2 + 3k + 1 + 2k + 2 \\
 &= k^3 + 3k^2 + 3k + 1 + 2k + 2
 \end{aligned}$$

2. Determine which of these are linear homogeneous recurrence relations with constant coefficients. Also, find the degree of those that are.

- a) $a_n = 3a_{n-2}$
- b) $a_n = 3$
- c) $a_n = a_{n-1}^2$
- d) $a_n = a_{n-1} + 2a_{n-3}$
- e) $a_n = a_{n-1}/n$
- f) $a_n = a_{n-1} + a_{n-2} + n + 3$
- g) $a_n = 4a_{n-2} + 5a_{n-4} + 9a_{n-7}$

$$a) \text{ Yes, Degree: } 2$$

$$b) \text{ Yes, Degree: } 0$$

$$c) \text{ No}$$

$$d) \text{ Yes, Degree: } 3$$

$$e) \text{ No}$$

$$f) \text{ No}$$

$$g) \text{ Yes, Degree: } 7$$

3. Solve these recurrence relations together with the initial conditions given.

- a) $a_n = 2a_{n-1}$ for $n \geq 1$, $a_0 = 3$
- b) $a_n = a_{n-1}$ for $n \geq 1$, $a_0 = 2$
- c) $a_n = 5a_{n-1} - 6a_{n-2}$ for $n \geq 2$, $a_0 = 1$, $a_1 = 0$
- d) $a_n = 4a_{n-1} - 4a_{n-2}$ for $n \geq 2$, $a_0 = 6$, $a_1 = 8$
- e) $a_n = -4a_{n-1} - 4a_{n-2}$ for $n \geq 2$, $a_0 = 0$, $a_1 = 1$
- f) $a_n = 4a_{n-2}$ for $n \geq 2$, $a_0 = 0$, $a_1 = 4$
- g) $a_n = a_{n-2}/4$ for $n \geq 2$, $a_0 = 1$, $a_1 = 0$

$$b) \quad a_n = 2$$

$$d) \quad r^2 - 4r + 4$$

$$r^2 - 4r + 4 = 0$$

$$\alpha_1 = 2, \alpha_2 = 2$$

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$$\alpha_1 = 2, \alpha_2 = 2$$

$$\alpha_1 = 2, \alpha_2 = 2$$

$$f) \quad r^2 - 4$$

$$r^2 - 4 = 0$$

$$\alpha_1 = 2, \alpha_2 = -2$$

$$\alpha_1 = 2, \alpha_2 = -2$$

$$\alpha_1 = 2, \alpha_2 = -2$$

$$\alpha_1 = 2, \alpha_2 = -2$$

$$\alpha_1 = 2, \alpha_2 = -2$$

$$\alpha_1 = 2, \alpha_2 = -2$$

11. The Lucas numbers satisfy the recurrence relation

$$L_n = L_{n-1} + L_{n-2},$$

and the initial conditions $L_0 = 2$ and $L_1 = 1$.

- a) Show that $L_n = f_{n-1} + f_{n+1}$ for $n = 2, 3, \dots$, where f_n is the n th Fibonacci number.
- b) Find an explicit formula for the Lucas numbers.

$$\begin{aligned}
 L_2 &= 3 \\
 f_n &= \{1, 1, 2, 3, 5, 8\}
 \end{aligned}$$

$$L_3 = 4$$

$$L_4 = 7$$

$$L_5 = 11$$

$$L_6 = 18$$

b)

$$r^2 - 1r - 1$$

$$\text{roots: } \frac{1}{2} \pm \frac{\sqrt{5}}{2}$$

$$\alpha_1 \left(\frac{1}{2} + \frac{\sqrt{5}}{2} \right)^n + \alpha_2 \left(\frac{1}{2} - \frac{\sqrt{5}}{2} \right)^n$$

$$\alpha_1 + \alpha_2 = 2 \quad \alpha_1 = 2 - \alpha_2$$

$$\frac{\alpha_1}{2} + \frac{\alpha_1 \sqrt{5}}{2} + \frac{\alpha_2}{2} - \frac{\alpha_2 \sqrt{5}}{2} = 1$$

$$\frac{\alpha_1 + \alpha_2}{2} + (\alpha_1 - \alpha_2) = 1$$

$$\frac{2 - \alpha_2 + \alpha_1}{2} + (2 - \alpha_2 - \alpha_1) = 1$$

$$3 - 2\alpha_1 = 1$$

$$\alpha_1 = 1$$

$$\alpha_2 = 1$$

$$\left(\frac{1}{2} + \frac{\sqrt{5}}{2} \right)^n + \left(\frac{1}{2} - \frac{\sqrt{5}}{2} \right)^n$$

13. Find the solution to $a_n = 7a_{n-2} + 6a_{n-3}$ with $a_0 = 9$, $a_1 = 10$, and $a_2 = 32$.

$$r^3 - 7r - 6$$

$$\text{Roots: } -1, -2, 3$$

$$\alpha_1(-1)^n + \alpha_2(-2)^n + \alpha_3 3^n$$

$$\alpha_1 + \alpha_2 + \alpha_3 = 9$$

$$-\alpha_1 - 2\alpha_2 + 3\alpha_3 = 10$$

$$\alpha_1 + 4\alpha_2 + 9\alpha_3 = 32$$

$$-\alpha_1 + 4\alpha_2 = 19$$

$$\alpha_2 = 4\alpha_1 - 19$$

$$9 - 4\alpha_1 + 29 - \alpha_3 + 16\alpha_1 - 76 + 9\alpha_3 = 32$$

$$-48 + 20\alpha_3 = 32$$

$$20\alpha_3 = 80$$

$$\alpha_3 = 4$$

$$\alpha_1 = 16 - 19 = -3$$

$$\alpha_1 - 3 + 4 = 9$$

$$\alpha_2 = 8$$

$$8(-1)^n - 3(-2)^n + 4 \cdot 3^n$$

20. Find the general form of the solutions of the recurrence relation $a_n = 8a_{n-2} - 16a_{n-4}$.

21. What is the general form of the solutions of a linear homogeneous recurrence relation if its characteristic equation has roots 1, 1, 1, 1, -2, -2, -2, 3, 3, -4?

20)

$$r^4 - 8r^2 + 16$$

$$\text{roots: } 2, -2$$

$$\alpha_1 1^n + \alpha_2 (-2)^n$$

21) $\alpha_1 + \alpha_2 n + \alpha_3 n^2 + \alpha_4 n^3 + \alpha_5 (-1)^n + \alpha_6 n(-2)^n + \alpha_7 n^2(-2)^n + \alpha_8 3^n + \alpha_9 n 3^n + \alpha_{10} (-4)^n$